Convolutional Neural Networks

CMSC 733 Fall 2015

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Overview

Goal: Understand what Convolutional Neural

Networks (ConvNets) are & intuition behind it.

1. Brief Motivation for Deep Learning 2. What are ConvNets?

3. ConvNets for Object Detection

First of all what is Deep Learning?

- Composition of non-linear transformation of the data.

- Goal: **Learn useful representations, aka features, directly from data.**

- Many varieties, can be unsupervised or supervised. ● Today is about ConvNets, which is a **supervised** deep learning method.

Recap: Supervised Learning

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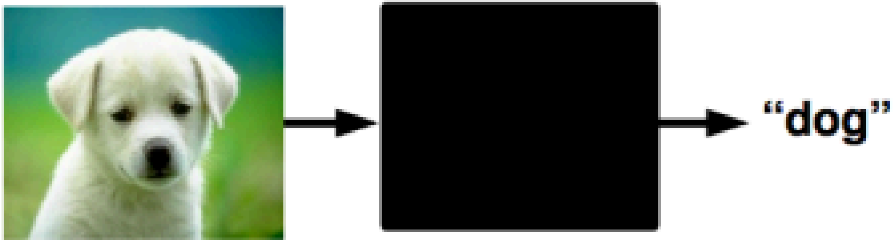
Supervised Learning: Examples

Classification



classification

Denoising

regression

OCR

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Supervised Deep Learning

So deep learning is about learning feature representation in a compositional manner.

But wait,

why learn features?

The Black Box in a

Traditional Recognition Approach

Feature
Extraction
(HOG, SIFT, etc)
Post-processing
(Feature selection,
MKL etc)
Classifier (SVM,
boosting, etc)

Preprocessing

The Black Box in a

Traditional Recognition Approach

Engineered Fe

ature

Classifier

Hand

Post-processing

Preprocessing
Extraction

(HOG, SIFT, etc)
(Feature selection,

MKL etc)
(SVM,

boosting, etc)

Extraction

(Feature selection,

Preprocessing

(HOG, SIFT, etc)

MKL etc)

Feature

Post-processing

- Most critical for accuracy
- Most time-consuming in development
- What is the best feature???
- What is next?? Keep on crafting better features? ⇒

Let's learn feature representation directly from data.

Learn features and classifier *together*

⇒ Learn an end-to-end recognition system. A non-linear map that takes raw pixels directly to labels.

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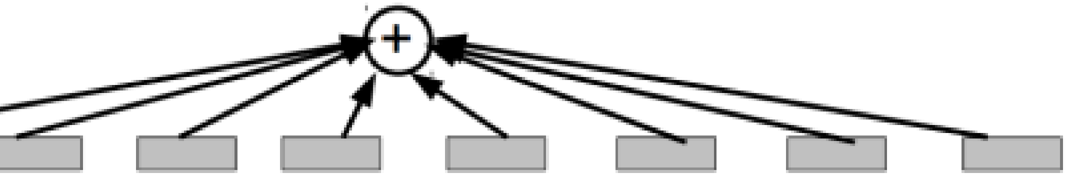
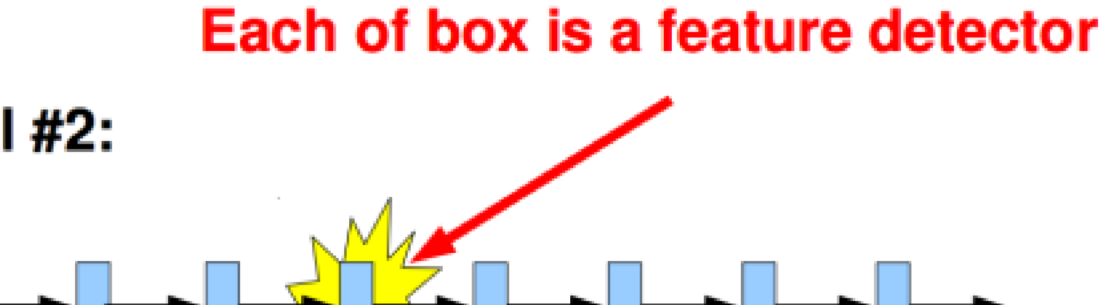
Building a

l #1:



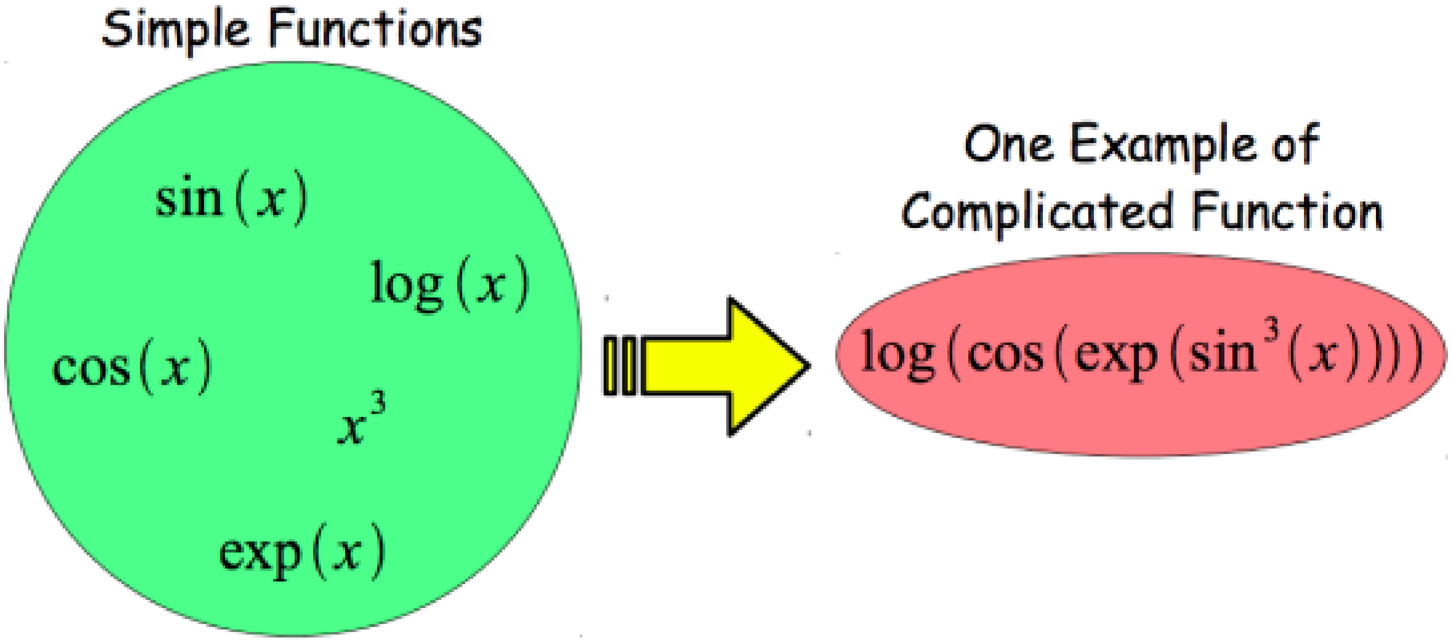
l #2:



complicated function Each box is a

simple nonlinear function

Building a complicated function



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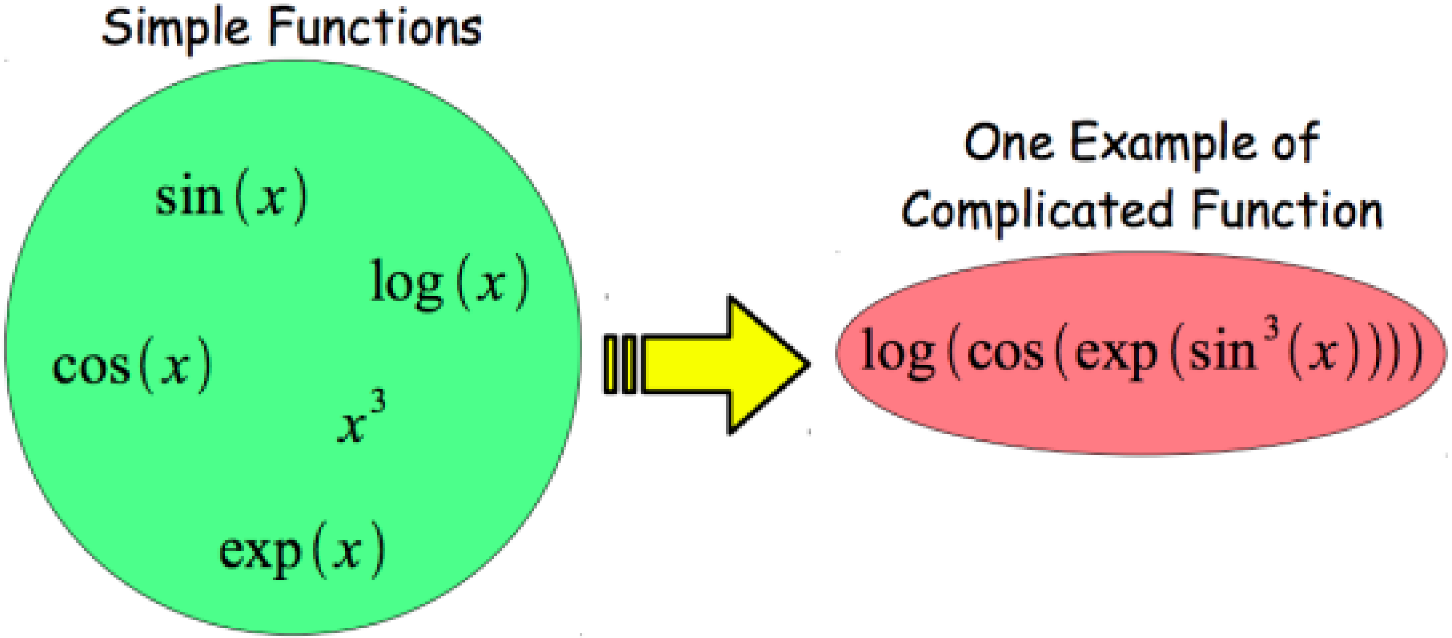

Building complicated functi

● Composition is at the core of deep learning methods ● Each “simple function” will have parameters subject to learning

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Intuition behind Deep Neural Nets


"CAR"

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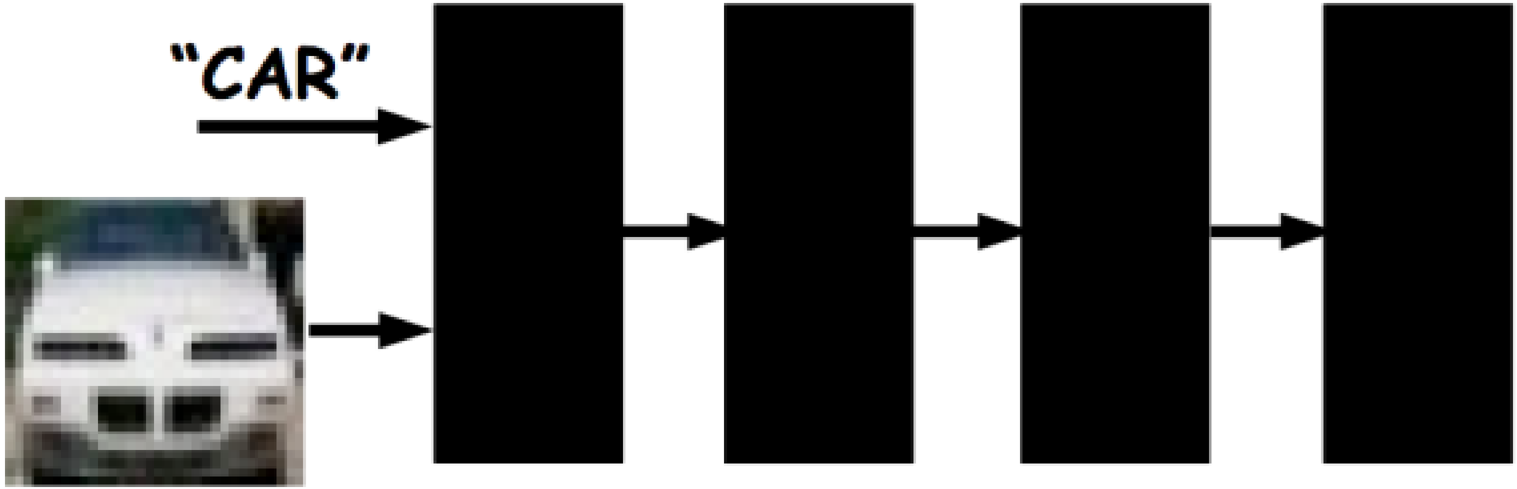
Intuition behind Deep Neural Nets



NOTE: Each black box can have trainable parameters.
Their composition makes a highly non-linear system.

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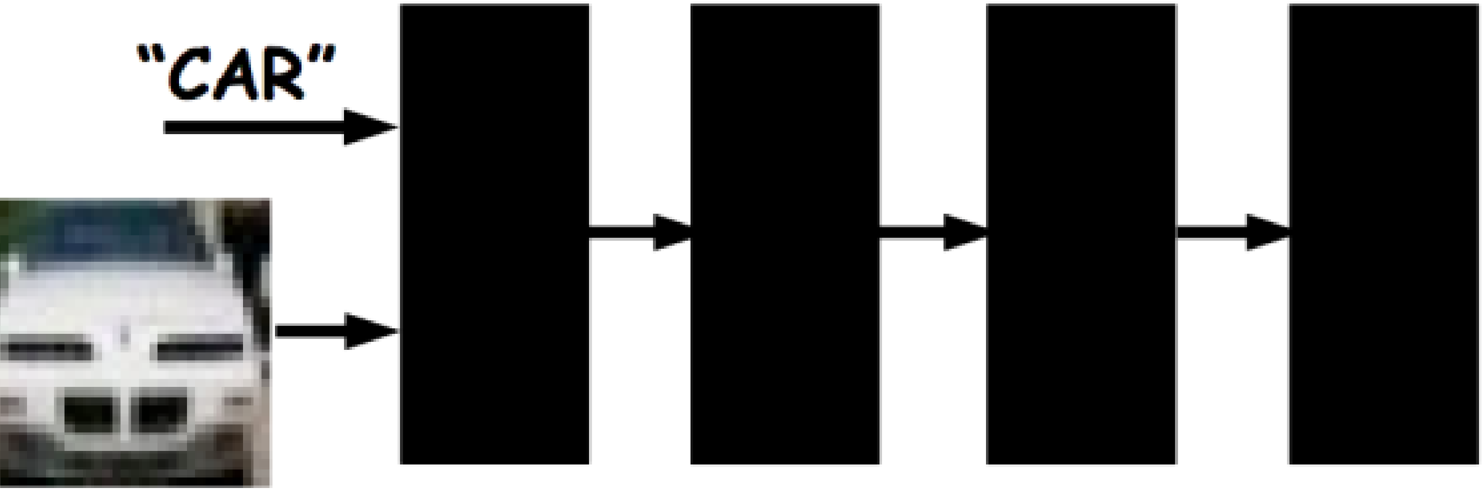
Intuition behind Deep Neural Nets


"CAR"

The final layer outputs a probability distribution of categories.

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A simple single layer Neural Network

Consists of a linear combination of input through a nonlinear function:

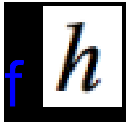
$$
\begin{aligned}
z &= Wx + b \\
a &= f(z)
\end{aligned}
$$

W is the weight parameter to be learned.

x is the output of the previous layer

f is a simple nonlinear function. Popular choice is max(x,0), called ReLu (Rectified Linear Unit)

1 layer: Graphical Representation



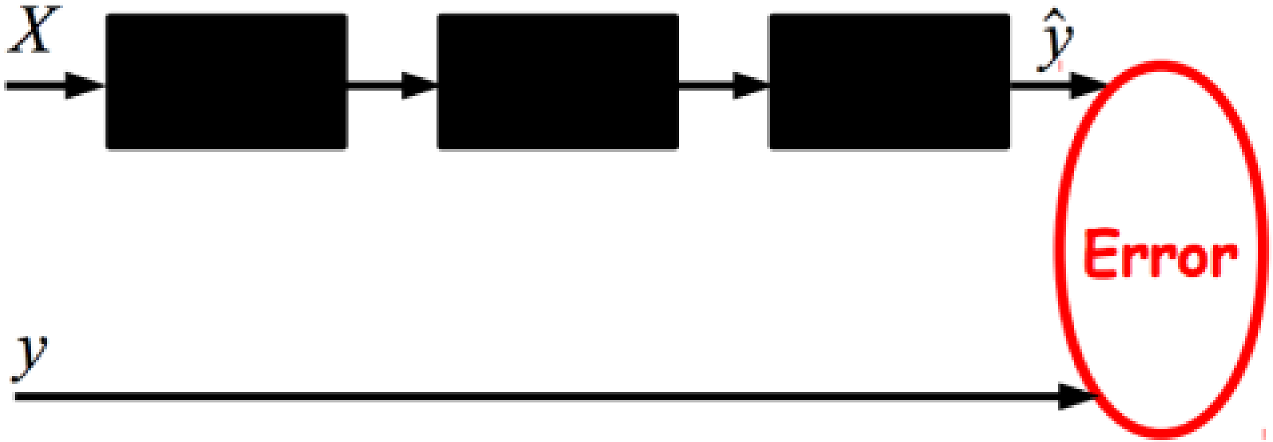
h is called a neuron, hidden unit or feature.

f

f

h

Joint training architecture overview



NOTE: Multi-layer neural nets with more than two layers are nowadays called **deep nets**!!

NOTE: User must specify number of layers, number of hidden units, type of layers and loss function.

Neural Net Training

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Neural Net Training

W.W W.

When the input data is an image..

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When the input data is an image..

- Reduce connection to local regions

Reuse the same kernel

everywhere

Because interesting features (edges) can happen at anywhere in the image.

Convolutional Neural Nets

Detail

If the input has 3 channels (R,G,B), 3 separate k by k filter is applied to each channel.

Output of convolving 1 feature is called a *feature map*.

This is just sliding window, ex. the output of one part filter of DPM is a feature map

Using multiple filters

Each filter detects features in the output of previous layer. So to capture different features, learn multiple filters.

Example of filtering

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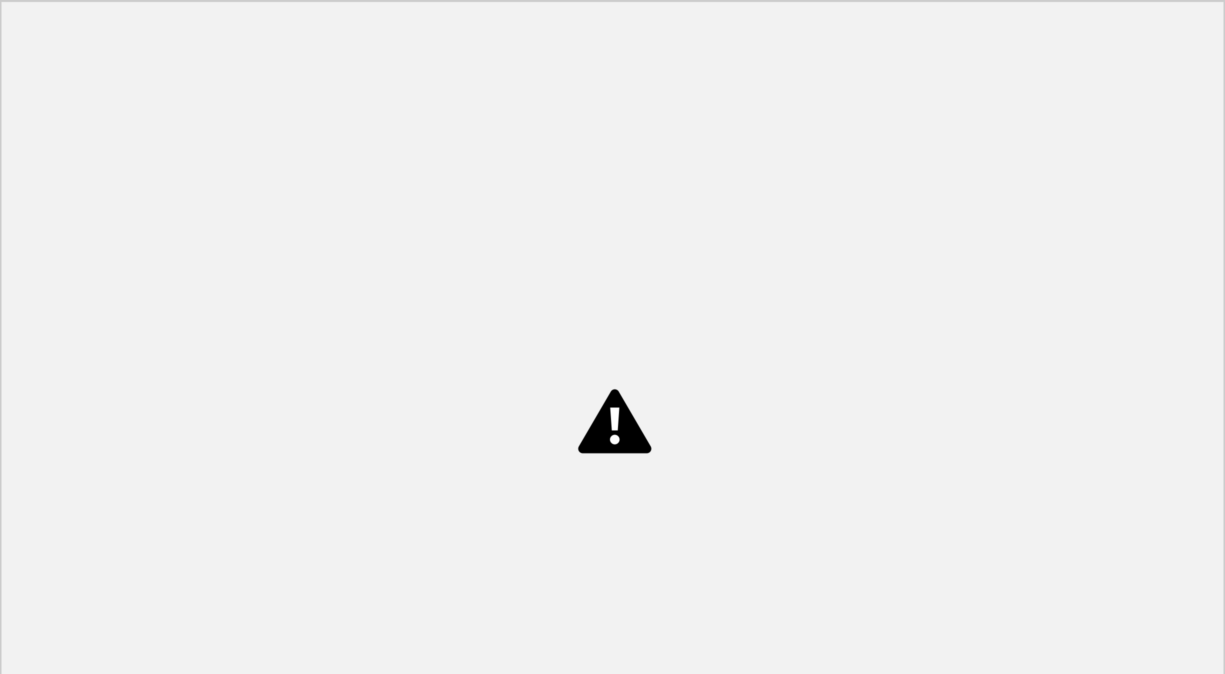
Building Translation Invariance

Building Translation Invariance via



Spatial Pooling

Pooling also subsamples the image, allowing the next layer to look at larger spatial regions.

Summary of a typical convolutional layer

Doing all of this consists one layer.

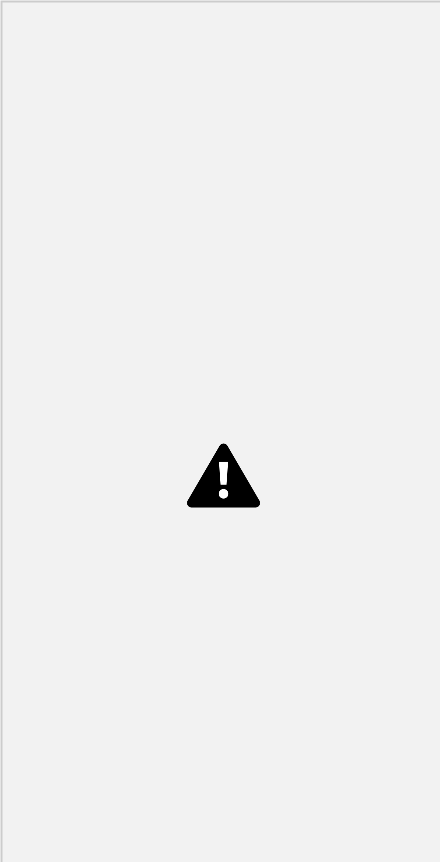
○ Pooling and normalization is optional.

Stack them up and train just like multi layer neural nets.

Final layer is usually fully connected neural net with output size == number of classes

Compare this to SIFT

Compare this to SIFT



Revisiting the composition idea

Every layer learns a feature detector by combining the output of the layer before. ⇒ More and more abstract features are learned as we stack layers.

Keep this in mind and let's look at what kind of things ConvNets learn.

Architecture of Alex Krizhevsky et al.

- 8 layers total.
- Trained on Imagenet Dataset (1000 categories, 1.2M training images, 150k test images)
- 18.2% top-5 error
    - Winner of the ILSVRC 2012 challenge.

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Architecture of Alex Krizhevsky et al.

First layer filters

Showing 81 filters of 11x11x3.

Capture low-level features like oriented edges, blobs.

Note these oriented edges are analogous to what SIFT uses to comute the radients.

Top 9 patches that activate each filter in

layer 1

Each 3x3 block shows
the top 9 patches for
one filter.

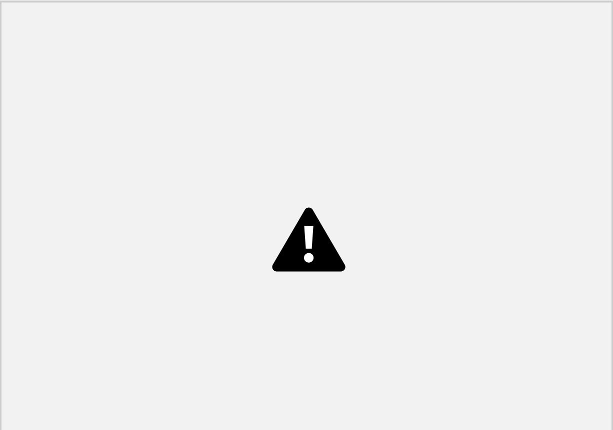
Second Layer

Second Layer

Note how the previous low-level features are combined to detect a little more abstract features like textures.

ConvNets as generic feature extractor

- A well-trained ConvNets is an excellent feature extractor.
- Chop the network at desired layer and use the output as a feature



representation to train a SVM on some other vision dataset.

- Improve further by taking a pre-trained ConvNet and re-training it on a different dataset. Called *fine-tuning*

One way to do detection with ConvNets

Since ConvNets extract discriminative features, one can crop images at the object bounding box and train a good SVM on each category. ⇒ Extract regions that are likely to have objects, then apply ConvNet + SVM on each and use the confidence to do maximum suppression.

R-CNN: Regions with CNN

features

Best performing method on PASCAL 2012

Detection improving previous methods by 30% [redacted] R. Girshick

ConvNet Libraries

• Cuda-convnet (Alex Krizhevsky, Google) • Caffe (Y. Jia, Berkeley, now Google) ● Pre-trained Krizhevsky model available. • Torch7 (Idiap, NYU, NEC) more around.